



GumBall6900

A signal-directed onchain fund

Fair-launched GBX, continuous sGBX signaling, market-executed acquisitions, and in-kind redemption.

● MINE

● SIGNAL

● ACQUIRE

● REDEEM

TARGET FINAL DESIGN

Pre-audit development concept. Not deployed. Not authorized for user funds.

PROTOCOL OVERVIEW

Abstract

GumBall6900 is a proposed onchain fund in which holders continuously direct new capital toward eligible assets. Anyone can mine GBX through a fixed public Fundraiser schedule. A holder may stake GBX one-for-one into non-transferable SignalGBX, ticker **sGBX**, and signal the active Strategies for assets they want the protocol to accumulate.

Incoming USDG follows current signals. A normal Strategy uses a bounded reverse Dutch auction to exchange that USDG for its target asset without requiring a price oracle in the core. After an acquisition, 90% of the received asset enters a shared Fund and 10% is streamed to eligible signalers. GBX can be burned to redeem a proportional, in-kind share of caller-selected Fund assets.

The result is not a token that tracks a predetermined index. It is a mechanism for forming a basket over time. Signals direct future flow rather than forcing the Fund to sell existing holdings. The intended final contracts are deployed once, are not upgradeable, and expose only four management actions.

THE CORE DISTINCTION

GBX relates to what the Fund already owns. sGBX directs what the Fund may acquire next.

IN PLAIN ENGLISH

Mine GBX publicly, signal the asset you want to accumulate, let new revenue follow live signals, and redeem GBX against selected assets already held by the Fund.

Status: This paper describes a target final design. The project is pre-audit, not deployed, and not authorized for user funds. The current development contracts still require reconciliation with this paper.

HOW TO READ THIS PAPER

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Start with the economic loop, then use the later sections for mechanics and risks.

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READING GUIDE

Pages 4-16 explain the design. Pages 17-19 state what must remain true, what can fail, and what is not finished.

FROM INDEX PRODUCT TO INDEX FORMATION

1. The problem

Onchain funds can move the basket onchain while leaving the basket-building process upstream.

Traditional funds make a basket easier to own, but investors normally receive a finished methodology. Membership, weighting, and rebalancing decisions are made elsewhere. Many onchain index products preserve the same basic structure: the token is onchain, but formation of the basket is still separate.

THE QUESTION

What if holders continuously directed where the fund's next dollar went?

TRADITIONAL BASKET

A methodology or committee decides membership and weights. Investors receive the finished result.

GUMBALL6900

Eligible Strategies define the available paths. Holders continuously decide which path receives new capital.

This changes the unit of coordination. Holders do not periodically approve a complete target portfolio. They signal among eligible acquisition paths, and every new unit of revenue follows the distribution that exists when it is routed.

The Fund records past acquisitions, while signals express present preferences. A changed signal affects future capital. It does not automatically sell assets already held.

IN PLAIN ENGLISH

The basket has a memory. Signals can change immediately, but the assets already acquired stay in the Fund until holders redeem them or market mechanics change them.

WHAT THE SYSTEM IS TRYING TO PRESERVE

2. Design goals

The design favors continuous direction, real asset custody, market execution, and a deliberately small control surface.

FAIR PUBLIC DISTRIBUTION Mine GBX through the Fundraiser without a team, founder, investor, presale, or advisor allocation.	CONTINUOUS DIRECTION Replace or reset sGBX allocations at any time; withdraw GBX after signals are cleared.
REAL ASSETS Hold raw tokens and redeem selected assets in kind rather than depend on a synthetic index number.	MARKET EXECUTION Use bounded reverse Dutch auctions so external participants decide when a price is acceptable.
GOVERNANCE MINIMIZATION Signals direct capital; management receives only four narrow actions; the core cannot be upgraded.	INSPECTABILITY Keep responsibilities narrow enough that money flow and failure boundaries can be reasoned about.

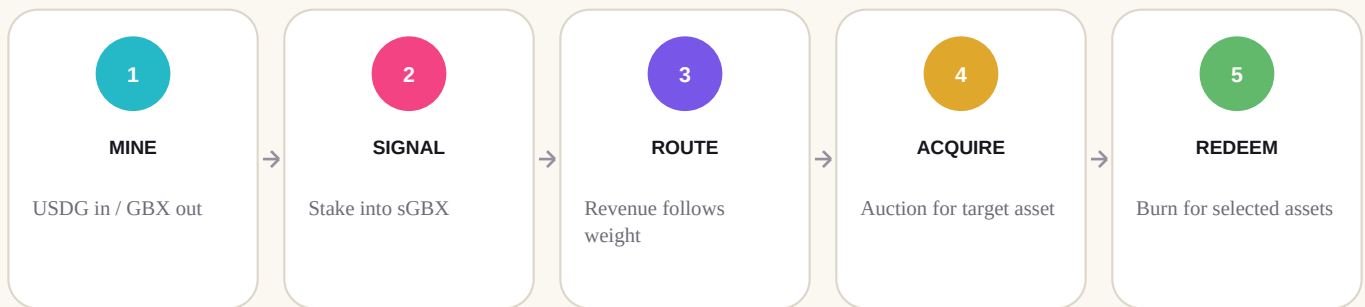
Non-goals

- No stable GBX price promise.
- No fixed portfolio composition or automatic rebalancing.
- No guarantee that an auction fills or a signal earns a reward.
- No protection from the risks of assets that enter the Fund.

THE ECONOMIC LOOP

3. System overview

The contract system has many boundaries; the holder-facing loop has five moves.



Every new distribution is another chance for current signals to shape the basket.

Normal revenue path

A USDG contribution enters through the Fundraiser, reaches Resonance, and is distributed among active Strategies according to current sGBX signal weight.

IN PLAIN ENGLISH

The routing layer does not decide what deserves capital. It reads the live signal distribution and applies it to the next revenue distribution.

No proposal season. No fixed signaling epoch. No withdrawal lock after allocations are reset.

SIX CONCEPTS ARE ENOUGH TO FOLLOW THE ECONOMICS

4. Protocol components

Implementation details stay behind narrow contract boundaries; the holder can reason through these six pieces.

USDG The stable asset contributed and routed toward acquisitions.	GBX The transferable protocol token with a one billion lifetime mint ceiling.
SIGNALGBX - SGBX Non-transferable staked GBX that measures live signal weight.	STRATEGY An eligible destination for USDG: normally an acquisition or buyback.
RESONANCE The allocation engine that distributes new USDG according to signals.	FUND The raw-token treasury supporting selected, in-kind redemption.

IN PLAIN ENGLISH

Money enters, signals direct it, Strategies acquire assets, the Fund holds most of them, and GBX provides the redemption claim.

Additional narrow contracts handle fundraising, canonical liquidity, Strategy and Bribe creation, auctions, and reward streaming. Those names matter for implementation review, but not for understanding the core loop.

FAIR LAUNCH THROUGH THE FUNDRAISER

5. GBX mining and supply

Mining is the familiar framing: contribute USDG through a public schedule and receive that period's GBX emission.

1,000,000,000 GBX

CUMULATIVE LIFETIME MINT CEILING



20M GENESIS LIQUIDITY - 2%

980M PUBLIC MINING CAPACITY - 98%

NO TEAM, FOUNDER, INVESTOR, PRESALE, OR ADVISOR ALLOCATION

Fixed declining schedule

The initial scheduled daily emission is **465,152.749681042811702004 GBX**. It is multiplied each day by **0.999525354337060160**, producing a 1,460-day, or four-year, half-life.

- Contributors within a day share that day's emission proportionally.
- An empty day forfeits its emission; nothing carries forward.
- Burning GBX never reopens mint capacity.
- Cumulative minting can never exceed one billion GBX.

TERMINOLOGY

Here, mining means contributing USDG through the Fundraiser. It does not mean proof-of-work.

WHY PRICE AND LIQUIDITY CREATE DEMAND FOR EMISSIONS

5.1 The mining market

A fixed GBX emission becomes an asset miners compete to acquire with USDG.

PRO-RATA DAILY MINING

$$GBX_i = \text{floor}(E_t \times \text{contribution}_i / \text{total_contributions}_t)$$

COMPETITIVE BENCHMARK: $\text{total_contributions}_t \approx GBX_price_t \times E_t$



Fund growth can reinforce GBX demand, but that feedback is not guaranteed.

WORKED EXAMPLE

100,000 GBX emitted x 0.50 USDG = 50,000 USDG gross emission value

If only 20,000 USDG has entered, the average gross cost is about 0.20 USDG per GBX before expenses and risk. That gap attracts competition. Near 50,000 USDG, the simple spot-price margin is mostly competed away.

WHAT THE CODE GUARANTEES

- Every contributed USDG enters the acquisition path.
- No team wallet sets a discretionary sale price.
- Successful normal acquisitions use the fixed 90/10 split.

WHAT THE MARKET CANNOT GUARANTEE

- Miners participate or GBX liquidity is deep enough.
- A quoted price is realizable after slippage and costs.
- Auctions clear or either GBX or Fund assets rise in value.

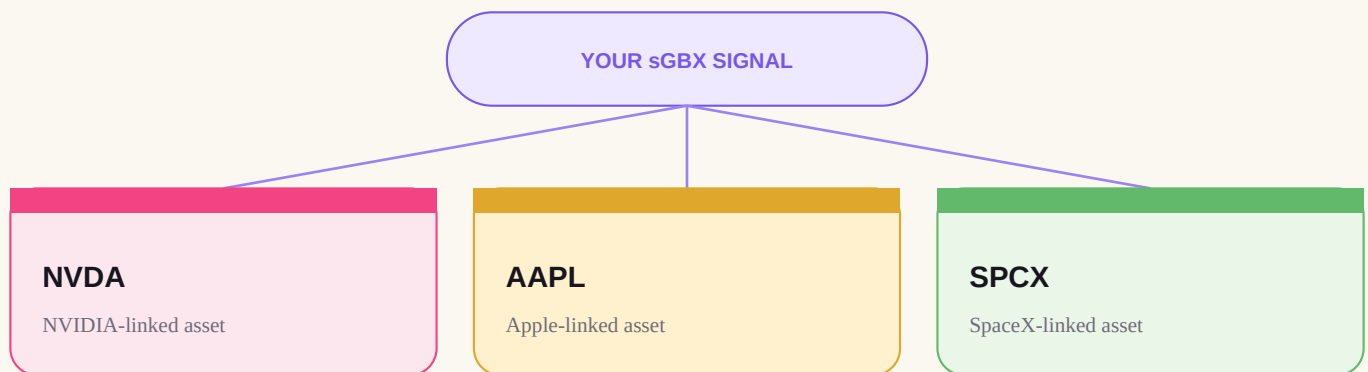
IN PLAIN ENGLISH

A usable GBX price gives each scheduled emission a USDG value. Rational miners can compete until the easy margin approaches break-even. That is a self-adjusting incentive - not an unconditional guarantee.

CHOOSE WHAT YOU WANT TO ACCUMULATE

6. Signaling with sGBX

Stake GBX one-for-one into non-transferable sGBX, then point that signal at active Strategies.



Signal the asset you want to accumulate; rewards still require completed acquisitions.

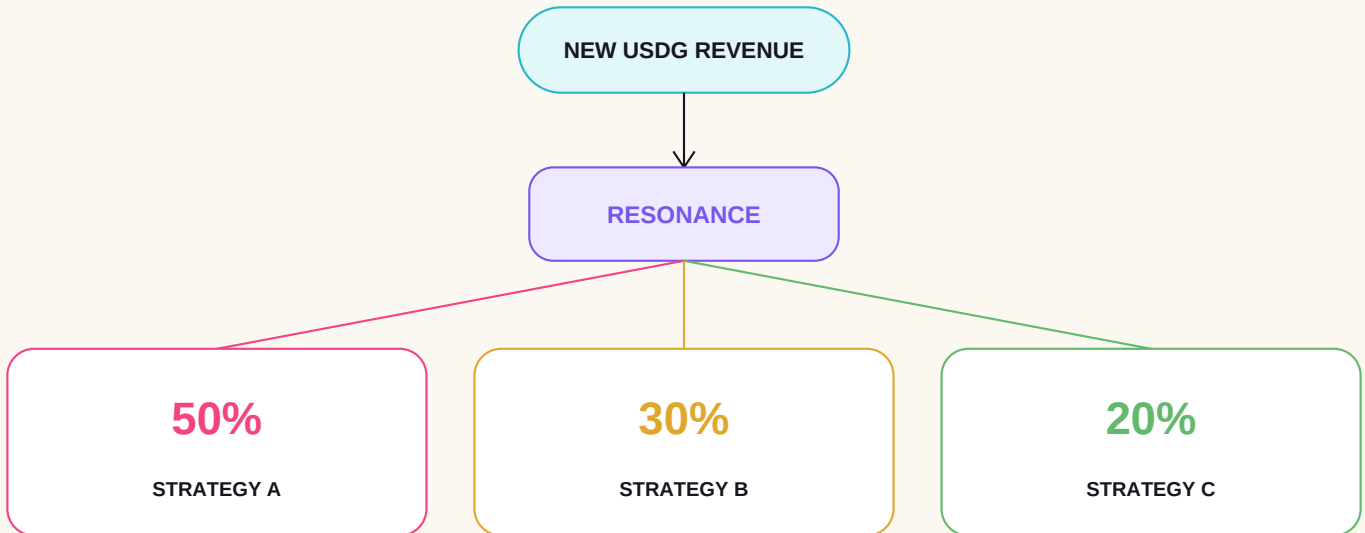
SIGNAL ANYTIME	REWARDS NEED FILLS	STRATEGY REQUIRED
Replace or reset allocations whenever your view changes.	A signal earns nothing unless the Strategy receives capital and completes an acquisition.	An asset can receive directed capital only after an eligible onchain asset has an active Strategy.

When a selected Strategy completes an acquisition, the signal-reward share is streamed in the acquired asset across eligible signal balances. A holder signaling NVDA can therefore earn the NVIDIA-linked asset; AAPL and SPCX work the same way when their Strategies are active.

THE NEXT DISTRIBUTION FOLLOWS LIVE WEIGHT

7. Capital allocation

Resonance applies the current signal proportions whenever it distributes newly received USDG.



CONCEPTUAL ALLOCATION

$$\text{allocation}_i = \text{floor}(\text{revenue} \times \text{signal}_i / \text{total_signal})$$

A Strategy holding 30% of active signal weight receives about 30% of the next distribution. Exact rounding and no-signal behavior must preserve value and avoid discretionary routing.

Signals apply to new flow. If preferences change after an acquisition, the Fund does not automatically rebalance. Its composition remains path-dependent: the result of every prior contribution, signal distribution, and completed acquisition.

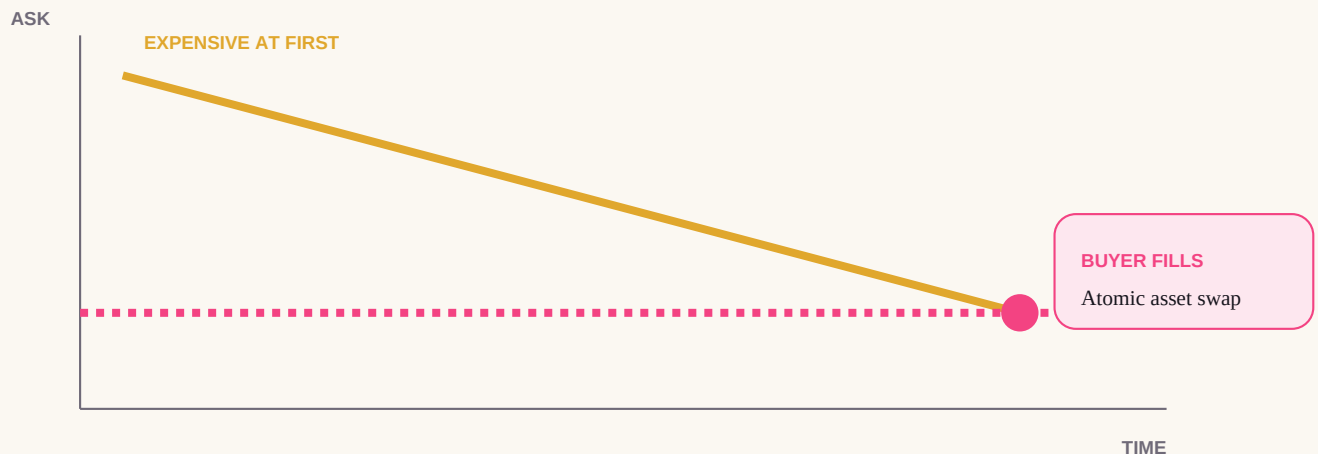
IN PLAIN ENGLISH

Moving your signal changes what happens next. It does not order the Fund to sell what it already owns.

BOUNDED REVERSE DUTCH AUCTIONS

8. Acquisitions

A Strategy offers accumulated USDG for a target asset while the amount requested from the buyer falls over time.

**1 - START HIGH**

The Strategy initially asks for an expensive amount of the target asset.

2 - PRICE FALLS

The required target-asset amount declines within configured bounds.

3 - BUYER FILLS

A market participant acts when the trade becomes acceptable.

4 - ATOMIC SWAP

USDG goes to the buyer and the target asset enters protocol flow in one transaction.

IN PLAIN ENGLISH

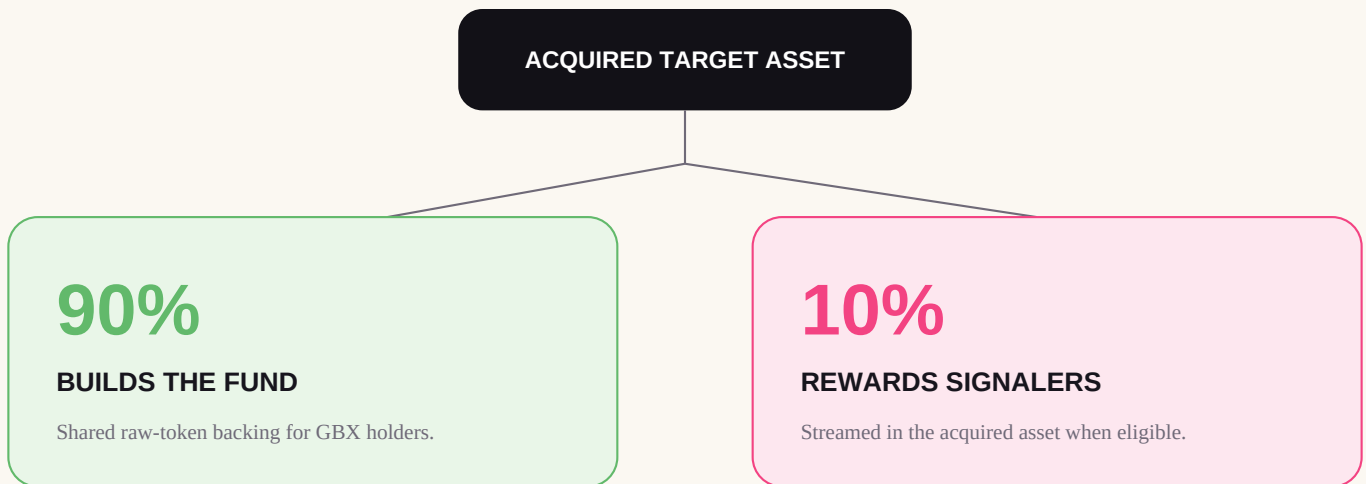
The market executes price discovery. The core does not need to maintain a price feed for every target asset.

This removes one oracle dependency but not market risk. Thin liquidity, unusual token behavior, poor participation, or badly chosen bounds can still delay execution or produce unfavorable results.

MOST BUILDS SHARED BACKING; SOME REWARDS THE SIGNAL

9. Fund formation and signal rewards

Every successful normal acquisition makes the holder's preference tangible while strengthening the common basket.



The 90/10 split is fixed before deployment and cannot be changed by the deployed core. If no eligible signalers exist, the signal-reward share returns to the Fund.

COMMON BENEFIT	PERSONAL INCENTIVE
Ninety percent becomes shared raw-token backing available through GBX redemption.	Ten percent lets eligible signalers earn the acquired asset they chose to support.

THE HOLDER EXPERIENCE

Signal NVDA and a completed NVDA acquisition can pay your share in the NVIDIA-linked asset, while the larger share joins the Fund for all GBX holders.

BURN GBX, CHOOSE ASSETS, RECEIVE YOUR SHARE

10. Fund behavior and redemption

The Fund holds raw tokens. The redeemer lists the Fund assets they actually want to claim.

PAYOUT FOR EACH SELECTED ASSET

$$\text{payout}_j = \text{floor}(\text{balance}_j \times \text{GBX_burned} / \text{supply_before_burn})$$

Example. Burn 1,000 GBX while total supply is 100,000 GBX: the redemption fraction is 1%. If the Fund holds 50,000 Asset X and 8,000 Asset Y, selecting both returns 500 X and 80 Y, subject to integer flooring.

PRE-BURN SNAPSHOTS Supply and selected balances are measured before GBX is burned.	ATOMIC EXECUTION The burn and every selected transfer succeed together or all revert.
CALLER SELECTION The holder may omit an unwanted or broken token.	PERMANENT OMISSION A claim for an omitted token stays with the remaining GBX supply.

The Fund is permissionless raw-token custody, not a curated registry. Official protocol membership is represented by active Strategies. Any ERC-20 sent directly to the Fund may become part of backing, even if it was never an official Strategy asset.

TWO SUPPLY-REDUCING LOOPS

11. Liquidity fees and buybacks

The canonical market position and buyback Strategies reinforce the system without creating a separate governance machine.

CANONICAL LIQUIDITY FEES**Collect without removing principal**

GBX fees -> burn

USDG fees -> signal flow

Fee collection is permissionless.

BUYBACK STRATEGY**USDG buys GBX and burns it**

100% of received GBX burned

No signal reward

GBX never sits as treasury inventory.

Canonical GBX/USDG position

The canonical position is precommitted, hookless, and funded with the 20 million GBX genesis allocation. Fee collection is permissionless and removes zero principal.

Buybacks

A buyback Strategy uses USDG to buy GBX and burns every received GBX atomically. It pays no signal reward and does not leave purchased GBX sitting as treasury inventory.

IN PLAIN ENGLISH

USDG fees return to holder-directed acquisitions. GBX fees and buyback purchases reduce supply through permanent burns.

SIGNALS GOVERN DIRECTION; THE CORE STAYS FIXED

12. Governance-minimized final design

Economic direction and system maintenance are deliberately separated.

HOLDER SIGNALS

Direct the flow

- Allocate sGBX
- Replace or reset anytime
- Shape future revenue

EXACTLY FOUR MANAGEMENT ACTIONS

Maintain the edges

- 1 ADD A STRATEGY
- 2 REMOVE A STRATEGY
- 3 CHANGE MANAGEMENT FEE
- 4 ADD BRIBE REWARDS

DEPLOYED ONCE - NO UPGRADES - NO MIGRATIONS - NO REWRITE

There is no general call executor, proxy upgrade, successor migration, arbitrary treasury withdrawal, pause function, signal-reward split setter, or general parameter setter.

The exact management-fee meaning, basis, and bounds remain to be specified in the final contracts. Strategy removal, Bribe reward registration, manager authorization, and key lifecycle must also be fully specified without creating a fifth management action.

THE TRADEOFF

Immutability reduces governance power, but it also makes mistakes permanent. That tradeoff is part of the design, not a claim that immutable code is automatically safe.

WHAT MUST REMAIN TRUE

13. Core invariants

The final implementation should turn each property below into explicit tests and reviewable contract behavior.

SUPPLY Cumulative minting never exceeds one billion; burns never restore capacity.	MINING Daily emissions follow the fixed sequential schedule; empty days forfeit emission.	ROUTING All contribution revenue enters the signal-directed allocation path.
SIGNALS Allocations may be replaced or reset without an epoch restriction.	REDEMPTION Pre-burn snapshots are used; the burn and selected transfers are atomic.	ACQUISITION A normal fill sends 90% to Fund and 10% to eligible signal rewards.
BUYBACK Every received GBX is burned atomically; no signal reward is paid.	LIQUIDITY Fee collection removes no principal, burns GBX fees, and routes USDG fees.	FIXED CORE No proxy upgrade or arbitrary asset-withdrawal path exists.

WHY INVARIANTS MATTER

An invariant is a rule that should still be true after every valid transaction, regardless of who calls it or in what order.

ONCHAIN DOES NOT MEAN RISK-FREE

14. Risks and trust assumptions

The design removes some discretionary powers, but it does not remove software, market, asset, or external-system risk.

SMART CONTRACTS A flaw can break routing, auctions, rewards, burns, or redemption. Immutability can make consequences permanent.	MANAGER KEY A compromised manager can misuse the four authorized actions even though it cannot rewrite the core.
SIGNAL CONCENTRATION Large sGBX holders may direct a disproportionate share of future capital.	MARKET EXECUTION Auctions depend on buyers, liquidity, bounds, and target-token behavior.
ASSET QUALITY Malicious, frozen, rebasing, or otherwise broken tokens may reach the Fund.	SELECTIVE REDEMPTION A broken selected token can revert the call; an omitted claim is permanently forfeited.
EXTERNAL SYSTEMS USDG, chains, bridges, sequencers, custody, and market infrastructure remain dependencies.	ECONOMIC AND LEGAL Demand, asset value, auction fills, jurisdiction, and tokenized-asset rights are not guaranteed.

RISK BOUNDARY

A smaller management surface narrows what an operator can do. It cannot guarantee good assets, good prices, good signals, or bug-free code.

A SPECIFICATION, NOT A LAUNCH CLAIM

15. Implementation status

This paper describes the intended final design. The repository remains a pre-audit engineering starting point.

1

FINAL INTERFACES

Four-action management surface

2

ECONOMICS

Management-fee meaning and bounds

3

VERIFICATION

Tests, models, and independent review

4

CONFIGURATION

Reviewed assets, chain, and deployment inputs

5

REHEARSAL

Reproducible deployment evidence

The current contracts, generated interfaces, indexing layer, deployment scripts, and technical documentation still describe a broader administrative surface. Those differences must be removed and the full repository reconciled before deployment.

CURRENT STATUS

A passing local build is engineering evidence only. It is not an audit, authorization, or launch claim.

Open design work includes the exact management-fee economics and bounds, final Strategy removal behavior, Bribe reward registration, manager authorization, key lifecycle, reviewed chain and asset inputs, deployment rehearsals, and independent security review.

A PUBLIC MECHANISM FOR FORMING A BASKET

16. Conclusion

GumBall6900 turns fund formation into a continuous onchain process rather than publishing a finished list.

GBX represents a transferable claim that can be redeemed against selected assets already held by the Fund. sGBX expresses where newly arriving capital should go. Strategies turn that signal into acquisitions, and the resulting history becomes the basket.

 GBX

A claim on selected assets already in the Fund.

 sGBX

A live direction for where new capital should go.

 FIXED CORE

The immutable rules both must follow.

THE PROPOSITION

Public mining. Continuous signals. Market-executed acquisitions. In-kind redemption. A fixed core.

The protocol does not promise that holders will select the best assets, that auctions will always clear, or that GBX will retain value. Its proposition is narrower: these mechanisms can make shared capital allocation more open and inspectable.

REFERENCE

Glossary and document basis

Short definitions for the terms used throughout this paper.

BRIBE The Strategy-specific reward contract that streams acquired assets across eligible signal balances.	FUND The raw-token treasury that holds acquired assets and supports selective in-kind redemption.	FUNDRAISER The public USDG contribution mechanism through which GBX is mined.
GBX The transferable protocol token with a one billion lifetime mint ceiling.	MANAGEMENT FEE The only adjustable fee in the target design; its definition and bounds are not yet fixed.	RESONANCE The allocation engine that distributes incoming USDG according to live sGBX signals.
SIGNALGBX - SGBX Non-transferable staked GBX used as live signal weight.	STRATEGY An active, eligible acquisition or buyback path.	USDG The stable asset contributed and routed through the protocol.

Document basis

This paper summarizes the intended target design recorded in the GumBall6900 repository as of 8 August 2026. The final reviewed contracts and their matching specifications remain the technical source of truth. If this paper and deployed bytecode ever disagree, the bytecode controls system behavior.

DISCLOSURE

Educational protocol overview only. Nothing in this document is investment, legal, or tax advice.